



User and Relevance Models & Social Media Analytics

IFN647 – Week 8

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Learning Objectives

Topic ID	Key topics	Mapping to slides
1	User Models	2-7
2	Relevance Models	8-13
3	Pseudo Relevance Models	14-22
4	Social Media Analytics	23-26
5	Social Search	27-34
6	Traditional IR Models vs. LLMs in Social Search	35-35
	References	36

1. User Models

- A substantial body of research has focused on learning **user profiles** to represent an individual's interests, enabling personalization in search and recommendation systems.
 - For example, queries can be expanded or reformulated using terms derived from a user's profile to better reflect their preferences.
- As discussed previously, **learning to rank** techniques can be employed to recommend relevant documents or items that align with a user's information needs and personalized interests.
- However, evaluating the outcomes of AI-based systems (e.g., conversational systems) remains challenging, as user interactions with results are complex and influenced by many factors.
- **User models** aim to characterize how users interact with ranked results, including what they examine, what they click, and how their behavior is affected by various biases (e.g., position bias or personal preferences).
 - Without incorporating user models, learning to rank methods may learn biased ranking functions, since observed user feedback (such as clicks) does not directly correspond to true relevance.
 - For instance, user models can be used to construct user profiles based on the content a user engages with, such as visited or clicked web pages, email communications, or documents created or edited on their device.
- Furthermore, user models play a critical role in interpreting implicit feedback: they help correct for biases in click data and enable more accurate estimation of true relevance from noisy observations.

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Ethical issues

- User models raise an important ethical concern for **Privacy and Data Collection**, such as **privacy violations** if data is collected without informed consent, or **unauthorized access**.
- Examples of ethical concerns
 - **Lack of control:** Users may not fully aware of what data is being collected and how it's being used.
 - **Lack of Consent:** Users may not always be informed about the extent of data collection or have a clear understanding of the privacy policy, e.g., how their information is being used to personalize their experience.
 - **Invasion of Privacy:** If the system tracks and analyzes user behavior, users may feel that their privacy is violated.
- User modelling and privacy issues do not mean that contextual information is useless, but rather that the benefits of any context-based approach need to be carefully examined.
- Other possible ML ethical issues can be occurred here, such as Bias and Fairness.

User Logs in User Models

- In general, user models may have limited effectiveness due to the challenges of constructing accurate user profiles.
 - Profiles are often **imprecise**, and users' information needs can **change dynamically over time**, reducing the reliability of static representations.
- However, there are important applications where contextual information significantly improves performance.
 - One notable example is the use of **query logs**, which provide rich contextual signals for understanding user intent.
- Query logs capture various forms of context, including:
 - Previous identical queries
 - Semantically similar queries
 - Query sessions, where multiple queries are issued within the same interaction context
- In addition, individual query histories can be utilized for applications such as caching or personalization mechanisms (e.g., via web cookies).

Privacy Issues in Query Logs

- A raw query log may have privacy issues. E.g., **User0003: "treatment for HIV symptoms"**, has the following privacy issues:
 - Reveals sensitive health-related concerns.
 - Could be linked back to an individual if combined with other data.
- Privacy-Preserving Approaches
 - To mitigate such risks, privacy-preserving techniques can be applied, such as:
 - Removing user identifiers (anonymization)
 - Generalizing query content to reduce sensitivity
 - E.g., remove user identifiers and generalize the raw log as
AnonUser: "treatment for medical condition"
- While query logs are highly valuable for enhancing user models and improving system performance, their use must be carefully balanced with privacy protection mechanisms to ensure ethical and responsible data handling.

Context vectors

- Context vectors are a widely used representation in user models for capturing a user's current interests, preferences, and interaction context in a structured numerical form.
- A common approach to constructing context vectors is through the use of **virtual documents**.
- A virtual document is an artificial document created by aggregating textual evidence from a user's interactions, such as previous queries, clicked web pages, and browsing history.
 - For example, one may select the top k terms (e.g., top 35 words) that frequently co-occur with a target term such as "*aquarium*". These co-occurring terms form a contextual representation of the concept.

zoology, cranmore, jouett, zoo, goldfish, fish, cannery, urchin, reptile, coral, animal, mollusk, marine, underwater, plankton, mussel, oceanography, mammal, species, exhibit, swim, biologist, cabrillo, saltwater, creature, reef, whale, oceanic, scuba, kelp, invertebrate, park, crustacean, wild, tropical

- This representation can then be used for query expansion. Specifically, candidate expansion terms (e.g., "*aquarium*") can be ranked with respect to a query (e.g., "*tropical fish*") by comparing their context vectors (or corresponding virtual documents).
 - The highest-ranked terms are selected as expansion terms to improve retrieval effectiveness.

Summary

- A common approach to user modeling is capturing query context via query logs, enabling analysis of word associations and identification of similar queries from click-through data, while ensuring privacy and careful data handling.
- When query logs are unavailable, context vectors or relevance models can serve as effective alternatives.
 - Expansion based on individual query terms (e.g., “tropical”) is often limited; considering the entire query context yields better results.
 - For example, using the Dice coefficient to retrieve virtual documents for the query “tropical fish” produces highly associated words such as:
goldfish, reptile, aquarium, coral, frog, exotic, stripe, regent, pet, wet
- LLM-based query expansion can further improve retrieval by incorporating contextually relevant terms, but it may sometimes generate irrelevant knowledge, potentially reducing search performance [4].

2. Relevance Models

- A **relevance model** characterizes the underlying topic distribution of documents that are relevant to a user's information need, often inferred from user feedback or observed interactions.
 - E.g., we can use a language modeling to represent **the topic for the relevance** as a probability distribution over terms (i.e., a relevance model).
 - In this framework, a query can be treated as a small sample of text generated from the relevance model, while relevant documents are considered larger, more informative samples.
- A relevance model can estimate the probabilities in **a given set of relevant documents** and then use this model to predict the relevance of new documents, or we can call it a Document Likelihood Model.
- Relevance models generate features for learning to rank, and user models provide personalized signals for relevance models incorporating user-specific context, such as preferences, history, and intent.
- User models can also be used to debias training data, particularly when relying on **implicit feedback** (e.g., clicks) or pseudo-relevance feedback.
 - This is achieved by explicitly modeling user behavior and associated biases, such as position bias and examination bias, through approaches like click derivation (as discussed in Week 6).
- By accounting for these biases, user models enable more accurate estimation of true relevance from noisy observations, leading to more reliable training signals for learning-to-rank.

Document likelihood model

- A simple document likelihood model $P(D|Q)$ uses the probability that a document D is relevant given the query Q .
- In contrast to the query likelihood model, $P(Q | D)$, which estimates the probability that a document generates a given query, the document likelihood model $P(D|Q)$ is more theoretical in the context of information filtering, because this approach is extensible to scenarios where no explicit query is available.
- User models can be used to capture user information needs (preferences) based on historical interactions (e.g., clicks, viewed documents), allowing the system to reformulate the problem as estimating $P(D|R)$, where R represents the relevance model (or user model).
- $P(D|R)$ is the probability of generating the text in a document (or selecting document) given a relevance model R . However, it has **two problems**:
 - 1) **Less effective than query likelihood due to difficulties comparing across documents (often documents are large) of different lengths. E.g., a document containing 10 words is very different to a document containing 1,000 words.**
 - 2) **We still have the problem of obtaining a large body of relevant documents.**

Kullback-Leibler (KL) Divergence

- It is an alternative approach to addressing **Problem 1)**
 - If we can estimate a relevance model (e.g., a language model) from a query or user feedback, we can compare this language model directly with a language model for a document.
 - Note that a document with a language model that is very similar to the relevance model is likely to be on the same topic.
- How to compare two language models (one for documents and another for the relevant documents or the relevance model)?
- A well-known measure from probability theory and information theory is the Kullback-Leibler (KL) divergence [1], which measures **the difference** between two probability distributions.

KL-Divergence cont.

- Given a true probability distribution P and an approximate distribution Q , the **negative KL divergence** serves as a measure of similarity, indicating how closely Q approximates P .

$$-KL(P||Q) = - \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

- Since KL- divergence is always positive (≥ 0 , $=0$ iff $P=Q$) and is larger for distributions that are further apart (not Convergence), we can use negative KL-divergence for ranking (measure the similarity between D and R), and
 - assume relevance model R for a topic is the true distribution (not symmetric), and the approximation to be the document language model (D):

$$\sum_{w \in V} P(w|R) \log P(w|D) - \sum_{w \in V} P(w|R) \log P(w|R)$$

Please note the second term of this equation does not depend on the document and can be ignored for the purpose of ranking.

Formula Derivation Explanation

$$\begin{aligned}
 -KL(R||D) &= - \sum_w P(w|R) \log \frac{P(w|R)}{P(w|D)} = \sum_w P(w|R) \log \frac{P(w|D)}{P(w|R)} = \\
 &\sum_w P(w|R) * (\log(P(w|D)) - \log(P(w|R))) = \\
 &\sum_w P(w|R) \log(P(w|D)) - \sum_w P(w|R) \log(P(w|R))
 \end{aligned}$$

- The second component is a constant for any document D, so we can ignore it if we use the equation to rank documents.

Example: KL-Divergence for query log matching

- Query Q “tropical fish”
- Query Log Q_i
 1. fish tank setup
 2. how to think clearly
 3. tropical fish types
 4. fish tank filter
 5. think tank meaning
 6. think best fish food
 7. clean fish tank
 8. think examples

- Formula
$$KL(Q||Q_i) = \sum_w P(w|Q) \log \frac{P(w|Q)}{P(w|Q_i)}$$
- Vocabulary for Q and Q_7 , $T = \{\text{tropical, fish, clean, tank}\}$
- Query language probabilities
 - $P(\text{tropical} | Q) = 0.5$,
 - $P(\text{fish} | Q) = 0.5$,
 - $P(\text{clean} | Q) = 0$
 - $P(\text{tank} | Q) = 0$
- Document language probabilities
 - $P(\text{tropical} | Q_7) = 0$,
 - $P(\text{fish} | Q_7) = 1/3$,
 - $P(\text{clean} | Q_7) = 1/3$
 - $P(\text{tank} | Q_7) = 1/3$

- Problem for using the formula:
 - $P(\text{tropical} | Q_7) = 0$
- How to solve it?
 - JM Smoothing
 - Assume $\lambda=0.25$
 - $P(\text{tropical} | Q_7) = (1-\lambda)*0 + \lambda*(1/25) = 0.01$
- $KL(Q||Q_7) = 0.5*\log(0.5/0.01) + 0.5*\log(0.5/0.333) = 2.16$

Example cont. KL-Divergence for query log matching

- Formula

$$KL(Q||q_i) = \sum_w P(w|Q) \log \frac{P(w|Q)}{P(w|Q_i)}$$

- Vocabulary for Q and Q_3 , $T = \{\text{tropical, fish, types}\}$
- Query language probabilities
 - $P(\text{tropical} | Q) = 0.5$,
 - $P(\text{fish} | Q) = 0.5$,
 - $P(\text{types} | Q) = 0$
- Document language probabilities
 - $P(\text{tropical} | Q_3) = 1/3$,
 - $P(\text{fish} | Q_3) = 1/3$,
 - $P(\text{types} | Q_3) = 1/3$
- $KL(Q||Q_3) = 0.5 * \log(0.5/0.333) + 0.5 * \log(0.5/0.333) = 0.405$
- Low KL value $\rightarrow$ close match, i.e., Q_3 is more similar to Q than Q_7 .

3. Pseudo-Relevance Model

- To address **Problem 2)**, we can collect user feedback on the results initially retrieved in response to a given query Q .
- **Pseudo relevance feedback** (blind feedback) is a method of finding an initial set of most likely relevant documents. Normally, we assume that the top " k " ranked documents are relevant to Q .
- Pseudo relevance feedback can be used to estimate a ***relevance model*** from **query Q** and **top- k ranked documents**.
 - Then we can rank new documents by similarity of document model to this *relevance model*.

Estimate probability distributions for relevance model R

- For a given user model R , based on pages 11 & 12, we have:

$$-KL(R||D) = \sum_{w \in V} P(w|R) \log P(w|D)$$

- The simple method is the maximum likelihood estimate for $P(w | R)$, based on the frequency in the query text, ranking score for a document is

$$\sum_{w \in V} \frac{f_{w,Q}}{|Q|} \log P(w|D)$$

- Where, Q can be expanded as **Q + top-k ranked documents**.
- $P(w | D)$ is rank-equivalent to query likelihood score (see week 5 lecture notes for the definition of $p(q_i | D)$ by using JM smoothing).
- Query likelihood model is a special case of retrieval based on relevance model.
- The above formulation can be updated by replacing Q with R if the user provides a set of relevant documents R .
- However, when no explicit set of relevant documents is available, pseudo-relevance feedback (PRF) can be applied. In this case, the top-ranked documents retrieved from an initial query are assumed to be pseudo-relevant and are used to **approximate R** .
- The query model is then refined based on these pseudo-relevant documents, typically by re-estimating term weights or expanding the query with terms extracted from them.

Estimating the Relevance Model

- We can also estimate $P(w|R)$ by the probability of pulling a word w out of the “bucket” representing the relevance model R depends on the n query words (terms) we have just pulled out

$$P(w|R) \approx P(w|q_1 \dots q_n)$$

- We view the probability of w is the conditional probability of observing w given that we just observed the query words $q_1 \dots q_n$
- By definition

$$P(w|R) \approx \frac{P(w, q_1 \dots q_n)}{P(q_1 \dots q_n)}$$

Estimating the Relevance Model cont.

- Joint probability is

$$P(w, q_1 \dots q_n) = \sum_{D \in \mathcal{C}} p(D) P(w, q_1 \dots q_n | D)$$

- Assume

$$P(w, q_1 \dots q_n | D) = P(w | D) \prod_{i=1}^n P(q_i | D)$$

- Gives

$$P(w, q_1 \dots q_n) = \sum_{D \in \mathcal{C}} P(D) P(w | D) \prod_{i=1}^n P(q_i | D)$$

- Where $\mathcal{C}$ is the set of top-k ranked documents; $P(D)$ is usually assumed to be uniform;
- $P(w, q_1 \dots q_n)$ is simply a weighted average of the language model probabilities for w in a set of documents, where the weights are the query likelihood scores for those documents
- Formal model for pseudo-relevance feedback
 - query expansion technique

Pseudo-Feedback Algorithm

Inputs: a document D , a query $Q = (q_1, \dots, q_n)$, a set of documents X , and a parameter k .

Output: the KL-divergence score for document D .

Method:

- 1) Let T be a set of indexing terms;
- 2) Rank documents in X using the query likelihood score for query Q ;
- 3) Select k top-ranked documents and add them into C ;
- 4) Calculate the relevance model probabilities $P(w|R)$ using the estimate for $P(w, q_1, \dots, q_n)$ for each term w in T ;
- 5) Rank documents in C again using the negative KL-divergence score

$$\sum_w P(w|R) \log P(w|D)$$

Example: Relevance Language Model R

- Query Q “fish tank”
- Query Log: Q_i
 1. fish tank setup
 2. how to think clearly
 3. tropical fish types
 4. fish tank filter
 5. think tank meaning
 6. think best fish food
 7. clean fish tank
 8. think examples

- Top rank results C
 - D_1 : fish tank setup
 - D_2 : fish tank filter
 - D_3 : clean fish tank
 - D_4 : tropical fish types

- Formula

$$P(w|R) = P(w, Q) =$$

$$P(w, q_1, \dots, q_n) =$$

$$\sum_{D \in C} P(D) P(w|D) \prod_{i=1}^n P(q_i|D)$$

- $T = \{\text{fish, tank, setup, filter, clean, tropical, types}\}$
- $P(\text{fish} | R) = P(\text{fish, fish, tank}) = P(D_1) * 1/3 * (1/3 * 1/3) + P(D_2) * 1/3 * (1/3 * 1/3) + P(D_3) * 1/3 * (1/3 * 1/3) + P(D_4) * 1/3 * (1/3 * 0) = 1/4 * 1/3 * (1/3 * 1/3) + 1/4 * 1/3 * (1/3 * 1/3) + 1/4 * 1/3 * (1/3 * 1/3) = 3 * (1/108) = 1/36$
- $P(\text{filter} | R) = P(\text{filter, fish, tank}) = 0 + P(D_2) * 1/3 * (1/3 * 1/3) + 0 + 0 = 1/4 * 1/3 * (1/3 * 1/3) = 1/108$
- $P(\text{tropical} | R) = P(\text{tropical, fish, tank}) = 0 + 0 + 0 + P(D_4) * 1/3 * (1/3 * 0) = 0$
- $P(\text{fish} | D_1) = 1/3$
- $P(\text{tank} | D_1) = 1/3$
- $P(\text{setup} | D_1) = 1/3$
- $P(\text{filter} | D_1) = 0$

Example of Relevance Language Model: the relevance term distribution and a document term distribution

Term w	$P(w, Q)$	$P(w R)$
fish	1/36	0.333
Tank	1/36	0.333
Setup	1/108	0.111
Filter	1/108	0.111
Clean	1/108	0.111
Tropical	0	0
Types	0	0

Term w	$P(w D_1)$
fish	0.333
Tank	0.333
Setup	0.333
Filter	0
Clean	0
Tropical	0
Types	0

Summary of Relevance Models

- Ranking based on relevance models requires two passes.
 - The first pass ranks documents using query likelihood to obtain the weights that are needed for relevance model estimation.
 - In the second pass, we use KL-divergence to rank documents by comparing the relevance model and the document model.
- Ranking by comparing a model of the query to a model of the document using KL-divergence is a generalization of query likelihood scoring.
- LLM-based Pseudo-Relevance Feedback CoT Prompt can be a promising method in the future.
- There is a continuous interaction between relevance models and user models
 - Relevance models generate ranked results
 - User interactions (clicks, dwell time) are observed
 - These interactions update the user model
 - The updated user model further refines the relevance model
 - This creates a dynamic, adaptive system

4. Social media analytics

- It is defined as, “the art and science of extracting valuable hidden insights from vast amounts of semi-structured and unstructured social media data (e.g., Twitter, Facebook) and users (e.g., user reviews and ratings) to enable informed and insightful decision making.”
- It is also commonly used by marketers to track online conversations about products and companies.
- There are three main steps in analysing social media
 - **Data identification**, identifying the subsets of available data to focus on for analysis;
 - **What content** is of interest. In addition to the text of content, we want to know: **who** wrote the text?
 - **Where** was it found or on which social media venue did it appear?
 - Are we **interested in information** from a specific locale?
 - **When** did someone say something in social media?
 - **Data analysis**, and
 - Information interpretation.

Social Media

Web-based services that allow individuals, communities, and organisations to produce, share and engage with user-generated content.

Media platforms and technologies

- e-commerce gateways;
- microblogs (e.g., Tumblr, Instagram, Twitter);
- social networking (e.g., LinkedIn, Facebook, MySpace);
- multimedia portals (e.g., Vimeo, Twitter, Facebook, Periscope, TikTok, YouTube);
- virtual worlds (e.g., Second Life);
- review platforms (e.g., Tripadvisor, Foursquare); and
- social gaming (e.g., World of Warcraft).

Microblog Retrieval

- Different types of microblogging technologies are available within social media to help achieve goals
- Twitter is a microblogging service introduced in March 2006. With over 125 million daily active users, Twitter is ranked among the most popular social media platforms.
 - The platform allows **everyone** to create and share information and ideas in real-time.
 - “Tweet” is a term that refers to a **short text message** that a Twitter user can produce.
 - This short plain text (tweet) can also include videos, photographs, and website URLs.
 - In the beginning, Twitter allowed 140 characters for a plain text message. The length has been expanded to 280 characters since Nov 2017.
 - The “@username” feature can be used to mention a specific person for interaction, e.g., user network
 - **Hashtag feature** can also be used to annotate user messages where the prefix “#” character is used as a non-spacing word, e.g., Hashtag based search.

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The screenshot shows a Google Scholar search for 'Microblog Retrieval' with 25 results. The left sidebar contains filters for 'Any time' (Since 2023, Since 2022, Since 2019, Custom range...), 'Sort by relevance' (Sort by date), 'Any type' (Review articles), and checkboxes for 'include patents' and 'include citations'. A 'Create alert' button is at the bottom. The main results list includes:

- An uncertainty-aware approach for exploratory microblog retrieval** by M.Liu, S.Liu, X.Zhu, Q.Liao, F.Wei... (IEEE transactions on ..., 2015 - IEEE explore.Ieee.org). It describes an uncertainty-aware microblog retrieval toolkit, MutualRanker, to estimate uncertainty introduced by the analysis algorithm as well as to quickly retrieve uncertainty-aware microblog retrieval using an ... (☆ Save Cite Cited by 71 Related articles All 8 versions).
- Short query expansion for microblog retrieval** by MA Zingla, L.Chiraz, Y.Slimani (Procedia Computer Science, 2016 - Elsevier). It details the use of proposed approaches for the microblog retrieval task of TREC. The experimental study was ... (☆ Save Cite Cited by 36 Related articles All 5 versions).
- Microblog Retrieval Using Topical Features and Query Expansion** by CH Lau, YF Li, D.Tjondronegoro (TREC, 2011 - academia.edu). It evaluates the performance of microblog retrieval based on standard Information Retrieval (IR) metrics, using a set of 50 topics provided by TREC. The topics are manu... (☆ Save Cite Cited by 58 Related articles All 7 versions).
- Query expansion for microblog retrieval** by ABandyopadhyay, K.Ghosh... (Journal of Web ..., 2012 - inderscienceonline.com). It discusses the extreme brevity of microblog posts (such as 'tweets') exacerbates the well-known vocabulary mismatch problem when retrieving tweets in response to user queries. In this study, we ... (☆ Save Cite Cited by 74 Related articles All 5 versions).
- Related searches**: microblog retrieval feedback, microblog retrieval disaster temporal microblog retrieval, microblog retrieval pseudo relevance feedback, microblog retrieval post disaster relief, microblog retrieval quality models, microblog retrieval ranking, microblog retrieval hashtag.
- Query expansion based on a feedback concept model for microblog retrieval** by Y.Wang, H.Huang, C.Feng... (of the 26th international conference on ..., 2017 - dl.acm.org). It tackles the problem of improving microblog retrieval al... search intent, which allows for microblog retrieval at a concept-level, ... model on the TREC Microblog corpora. The experimental ... (☆ Save Cite Cited by 57 Related articles All 3 versions).
- Language processing for arabic microblog retrieval** by K.Danwishi, W.Magdy, A.Mourad (Proceedings of the 21st ACM ..., 2012 - dl.acm.org). It illustrates some of the challenges associated with Arabic microblog retrieval, ... We present some of the required processing for effective retrieval such as improved ... (☆ Save Cite Cited by 99 Related articles All 2 versions).
- Quality models for microblog retrieval** by J.Choi, WB.Croft, JY.Kim (Proceedings of the 21st ACM international ..., 2012 - dl.acm.org). It states that a microblog is a medium similar to a traditional blog, ... blog in that the users of microblogs tend to post smaller content ... In this paper, we focus on microblog ad-hoc retrieval, which ... (☆ Save Cite Cited by 44 Related articles All 3 versions).
- Information search and retrieval in microblogs** by M.Efron (Journal of the American Society for Information ..., 2011 - Wiley Online Library). It is posed against an index of microblog data. Based on some query or a document in microblog retrieval systems is quite fluid, ... search process in the context of retrieving microblog data. ... (☆ Save Cite Cited by 201 Related articles All 16 versions).
- Microblog retrieval for post-disaster relief: Applying and comparing neural ir models** by P.Khosla, M.Basu, K.Ghosh, S.Ghosh (arXiv preprint arXiv:1707.06112, 2017 - arxiv.org). It states that retrieval are needed for various applications. In this work, we experiment with microblog retrieval ... In this work, we experiment with a number of neural network based retrieval models, ... (☆ Save Cite Cited by 25 Related articles All 3 versions).
- Effective pseudo-relevance for microblog retrieval** by K.Albishi, Y.Li, Y.Xu... (of the Australasian computer science week ..., 2017 - dl.acm.org). It states that retrieving relevant documents from microblogs is somewhat challenging because of its nature and the massive scale of the data. However, microblog retrieval ... microblog retrieval ... (☆ Save Cite Cited by 20 Related articles All 4 versions).

Query likelihood

- It uses Bayesian (Dirichlet prior) smoothing. It considers both the document and the query size.

$$P(Q, d) = \sum_{w \in Q, d} c(w, Q) \log \left(1 + \frac{c(w, d)}{\mu \times P(w|C)} \right) + |Q| \log \frac{\mu}{\mu + |d|}$$

where

- $c(w, d)$ is the word count in the given document d .
- $c(w, Q)$ is the word count in the given query Q .
- $|d|$ and $|Q|$ are the respective lengths (size) of the document and query.
- $P(w|C)$ is the probability of the word in the collection C that is used to normalise the model.
- μ is the smoothing parameter.

5. Social Search

Social search

Communities of users *actively participating* in the search process
Goes beyond classical search tasks

Key differences

- Users interact with the system
- Users interact with other users either implicitly or explicitly
- User generated content
- Users can tag their own and other's content

User Tags

- Then: Library card catalogs
 - Indexing terms chosen with search in mind
 - Experts generate indexing terms
 - Terms are very high quality
 - Terms chosen from controlled vocabulary
- Now: Social media tagging
 - Tags not always chosen with search in mind
 - Users generate tags
 - Tags can be noisy or even incorrect
 - Tags chosen from *folksonomies* <https://en.wikipedia.org/wiki/Folksonomy>
 - A Folksonomy is a classification system
 - The collective assemblage of tags assigned by many users
 - Make the use of public tags effective.

Types of User Tags

- Content-based
 - car, woman, sky
- Context-based
 - new york city, empire state building
- Attribute
 - nikon (type of camera), black and white (type of movie), homepage (type of web page)
- Subjective
 - pretty, amazing, awesome
- Organizational
 - to do, my pictures, readme

Taxonomy vs Folksonomy

- | | |
|--|---|
| <ul style="list-style-type: none"> • Hierarchical <ul style="list-style-type: none"> – Parent/child & sibling relationship • Exclusive <ul style="list-style-type: none"> – The same item can not be in two distinct categories • Top-down <ul style="list-style-type: none"> – Established by an authority | <ul style="list-style-type: none"> • Flat <ul style="list-style-type: none"> – No levels, order, explicit relationship • Not Exclusive <ul style="list-style-type: none"> – An item can be associated to many tags • Bottom-up <ul style="list-style-type: none"> – Created by users |
|--|---|

Example of Folksonomy

Mendeley is most known for its reference manager to manage and share research papers and generate bibliographies for scholarly articles.



Searching Tags

- Tags can be used to describe textual or non-textual items (e.g., images or videos) to provide a textual dimension to items.
- These textual representations of items can be very useful for searching; however, tags are **very sparse representations** of very complex items.
- Searching user tags is challenging
 - Most items have only a few tags
 - Tags are very short
- Boolean, probabilistic, vector space, and language modeling will fail if use naïvely.
- Must overcome the ***vocabulary mismatch problem*** between the query and tags.
- Possible ways to overcome this problem
 - 1) Stemming (e.g., stem classes)
 - 2) Pseudo-relevance feedback for tag expansion

Tag Expansion

- It uses search results (pseudo-relevance feedback) to enrich a tag representation.
- It overcomes vocabulary mismatch problem by expanding tag representation with external knowledge.
- Possible external sources
 - Thesaurus
 - Web search results
 - Query logs
- After tags have been expanded, we can use standard retrieval models.

Tag Expansion Procedure

- (1) Use tag “**tropical fish**” as a query Q to find top-k results C ;
- (2) Select terms w with the highest probability $P(w|Q)$, e.g., terms “fish”, “tropical”, “aquariums”, “goldfish”, and “bowls”;
- (3) Q is expanded as $Q' =$ “fish, tropical, aquariums, goldfish, bowls”;
- (4) Search by using the enriched query Q' .

• **Question:** How to calculate $P(w|Q)$?

• **Answer:**

- According to the estimating relevance model (page 17), we assume $R \approx Q$.
- $P(w|Q) = P(w | \text{“tropical fish”}) \propto P(w, \text{‘tropical’, ‘fish’}) =$
- $\sum_{d \in C} P(d) P(w|d) \prod_{i=1}^n P(q_i|d) =$
- $\sum_{d \in C} P(d) P(w|d) P(\text{‘tropical’}|d) P(\text{‘fish’}|d)$
- where $P(x|d)$ is a query likelihood score (review week 5 JM smoothing)

$$p(q_i|D) = (1 - \lambda) \frac{f_{q_i,D}}{|D|} + \lambda \frac{c_{q_i}}{|C|}$$

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Example 2.

32

Age of Aquariums - Tropical Fish

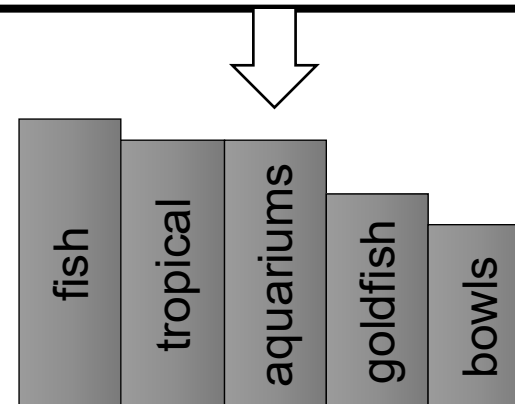
Huge educational aquarium site for **tropical fish** hobbyists, promoting responsible **fish** keeping internationally since 1997.

The Krib (Aquaria and Tropical Fish)

This site contains information about **tropical fish** aquariums, including archived usenet postings and e-mail discussions, along with new ...

...

Keeping Tropical Fish and Goldfish in Aquariums, Fish Bowls, and ...
Keeping **Tropical Fish** and Goldfish in Aquariums, **Fish** Bowls, and Ponds at AquariumFish.net.



$P(w | \text{“tropical fish”})$

Inferring Missing Tags

Issues in Searching Tags

- Even with tag expansion, searching tags is challenging.
 - Tags are inherently noisy and incorrect.
 - Many items may not even be tagged!
 - Typically, it is easier to find popular items with many tags than less popular items with **few/no tags**.
-
- As we just described, items that have no tags pose a challenge to a search system.
 - How can we automatically tag items with few or no tags?
 - Uses of inferred tags
 - Improved tag search
 - Automatic tag suggestions

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Methods for Inferring Tags

- TF*IDF if **items are textual**, such as books, or news articles.

$$wt(w) = \log(f_{w,D} + 1) \log\left(\frac{N}{df_w}\right)$$

- where $f_{w,D}$ is the number of times term w (tag) occurs in item D , N is the total number of items, and df_w is the number of items that term w occurs in.
- Classification
 - Train binary classifier for each tag (use all of the existing tag/item pairs as training data to train the classifiers, and **represent an item as a feature vector**)
 - Performs well for popular tags, but not as well for rare tags.
- Maximal marginal relevance
 - Finds tags that are relevant to the item and **novel with respect to existing tags** (or not very similar to any of the other tags), where t is tag, i is an item and T_i is the current set of tags for item i .

$$MMR(t; T_i) = \left(\lambda Sim_{item}(t, i) - (1 - \lambda) \max_{t \in T_i} Sim_{tag}(t_i, t) \right)$$

6. Do LLMs Really Outperform Traditional IR Models in Social Search?

1. **Ask** a social search engine
<https://www.ask.com/>

2. Ask **ChatGPT**, by using the following prompt:

A user asks: “Do LLMs really outperform traditional retrieval models in social search?”

Search relevant social media discussions and extract evidence-based arguments, examples, and counterexamples mentioned by users. Provide a balanced answer grounded in the retrieved posts.

3. Opinionated LLMs that align with or support the user’s views can exacerbate polarization, creating a risk of “generative echo chambers” [5].

4. With millions using LLM-powered tools, these limitations raise concerns about **opinion influence**, the spread of biased information, and the need for regulation and technical safeguards [5].

” Ask ChatGPT

Synthesized Answer (Grounded in Social Discussions)

Social media discussions suggest that LLMs **do not universally outperform traditional retrieval models** in social search; instead, their effectiveness depends strongly on the task, scale, and evaluation criteria. Users consistently highlight **clear strengths, well-known limitations, and practical trade-offs**.

Arguments Supporting LLMs

Many practitioners and researchers argue that LLMs **outperform traditional retrieval models** (e.g., BM25, TF-IDF) in scenarios where **semantic understanding and context aggregation** are critical.

Evidence and examples mentioned by users:

- Users report that LLM-based systems retrieve **semantically relevant posts** even when there is little lexical overlap (e.g., slang, paraphrases, abbreviations common in social media).
- Several posts describe LLMs as being especially effective for:
 - Opinion mining and stance detection
 - Multi-turn or conversational social search
 - Queries like “What do people think about X?” rather than “Find posts containing keyword X”
- Researchers note that LLMs can **summarize diverse viewpoints** across many posts, which traditional rankers cannot do directly.

Typical sentiment:

“BM25 finds posts; LLMs find *meanings*.”

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Do LLMs Really Outperform Traditional IR Models in Social Search?

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llmrefs.com

<https://www.llmrefs.com/>

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Reference

www.reference.com/business-finance/llms-dramatic-effect-evolution-ai-companies

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Reference

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Faqtoids

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The Surprising Ways LLMs Are Reshaping Artificial Intelligence

Large Language Models (LLMs) have emerged as transformative forces within the artificial intelligence landscape. Their profound impact is not just technical but strategic, reshaping...

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